Trainings

Select Service Tools

Recommended Cummins® Service Tools

- Spray cleaner, Part Number 3823717
- Chip removing unit, Part Number 3164019
- Sealant, Part Number 3375068

Additional Service Items

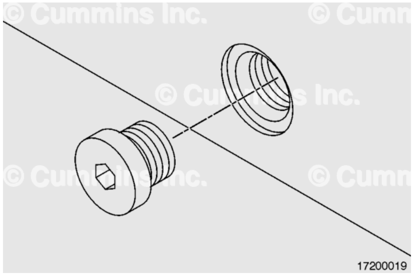
- Allen wrench or socket

General Information

This procedure addresses repair of threaded holes using different methods while highlighting strengths and weaknesses of each method. Threaded holes on critical components, such as crankshafts and connecting rods, should **not** be repaired.

Remove

Select the appropriate size Allen wrench or socket, and remove the plug.

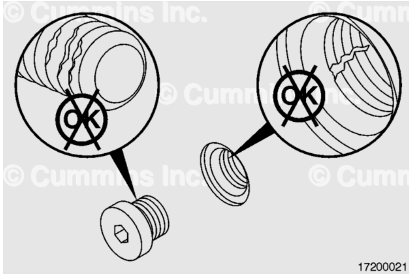


Clean and Inspect for Reuse

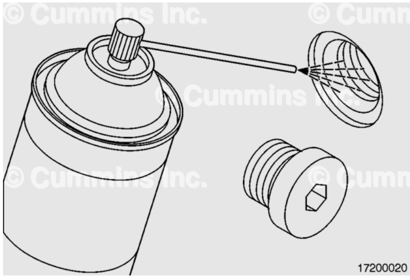
Inspect the threads of the pipe plugs for mutilation or other damage.

Inspect the threaded bores for damage.

Repair the bores, if necessary.



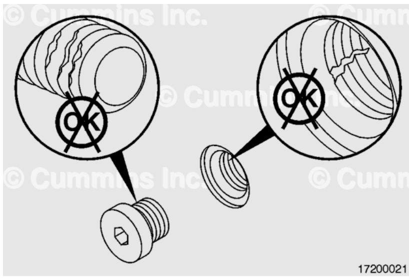
Use spray cleaner, Part Number 3823717 or equivalent, to clean the threads of the straight thread plugs and threaded bores.



Inspect the threads of the pipe plugs for mutilation or damage.

Inspect the threaded bores for other damage.

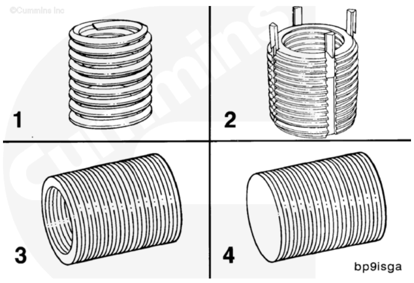
If the threaded bores are damaged, repair the threads according to the Repair section in this procedure.



Repair

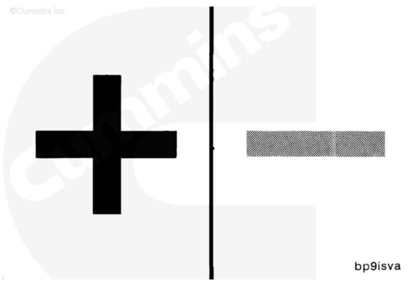
Four different types of thread inserts can be used to repair damaged capscrew threads:

1. Coiled thread inserts
2. Keylock thread inserts
3. Open end thread inserts
4. Closed end (blind) thread inserts.



Coiled thread repair inserts have the following advantages:

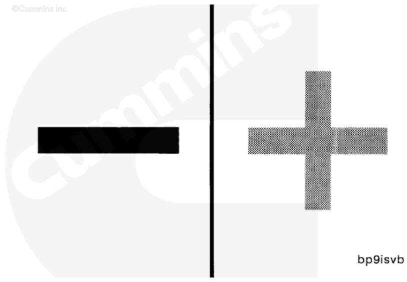
- Can be installed in locations that have minimal thickness around the threaded hole
- Made of materials that resist and prevent galvanic action
- Sealants are **not** required when installing repair inserts



- Equivalent to or exceeds the strength of the original threads
- Finish machining is **not** required.

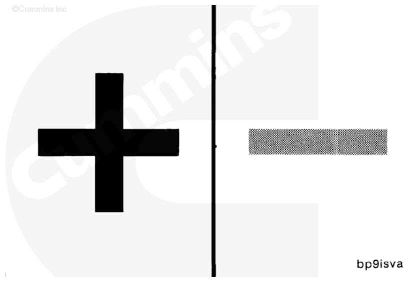
Coiled thread repair inserts have the following disadvantages:

- Special sized tap drills or reamers are required
- Special taps are required
- Special installation tools are required
- The repair insert can **not** be reused if it "backs out" after installation
- The coiled thread insert has no bottom and provides no sealing capabilities.



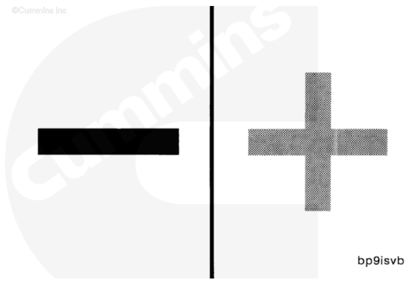
Keylock thread inserts have the following advantages:

- Standard, commercially available, drills and taps are used during installation
- Relatively inexpensive and available from several sources
- Can be removed and replaced without requiring modification or rework to the component.



Keylock thread inserts have the following disadvantages:

- Generally apply to repair of components that have thicker wall sections
- Installation requires a locking key driver.



Note : Inserts can be used in cast iron, steel, copper, brass, bronze, aluminum, magnesium, plastic, and wood.

For special installations, call or write the following manufacturers.

Perma-Thread™	Microdot Inserts 1025 N. Amanda Street Anaheim, CA 92806 (714) 870-6650
Heli-Coil™	Heli-Coil™ Products Shelter Rock Lane Danbury, CT 06810 (203) 743-7651

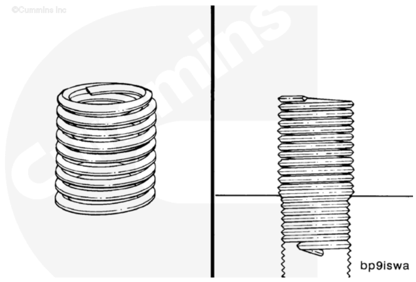
Coil Thread	Tridair Industries 3000 West Lomita Blvd. Torrance, CA 90505 (213) 530-2220

Note : This part of the procedure is for repairing coiled thread inserts.

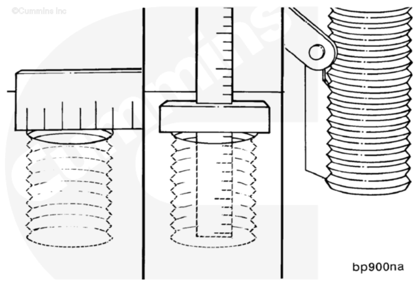
This repair procedure provides useful information and installation instructions for coiled thread inserts. Coiled thread inserts are a precision wound wire insert for repairing failed or mutilated internal threads.

These coils in their free state are designed to be larger in diameter than the tapped hole. After installation, the insert conforms to the outer wall of the tapped hole. This compressed fit holds the insert in place.

Note : Inserts are normally made of stainless steel wire. However, other materials are available to meet special installation requirements.

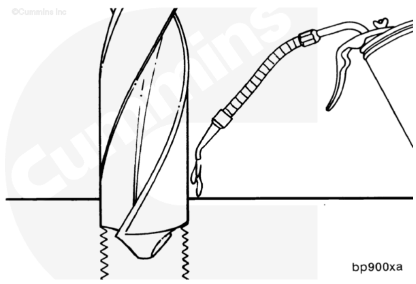


Determine the diameter, depth, and pitch of the original threaded hole.



⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear protective goggles when drilling metal.



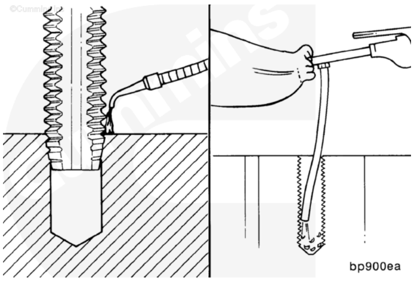
Use cutting oil while drilling the capscrew hole to the specified diameter. Use the charts in this procedure for drill size diameter specifications. Drill to the bottom of blind capscrew holes.

Use the special tap specified in the chart, according to the thread size needed, to cut the threads in the capscrew hole.

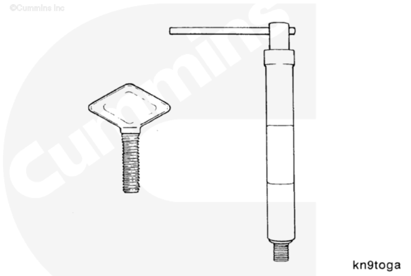
Note : Use cutting oil when tapping the capscrew hole. Clean the capscrew hole before installing the repair

insert.

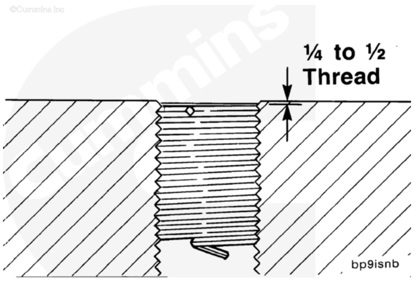
Use chip removing unit, Part Number 3164019, to remove machining chips from the capscrew hole.



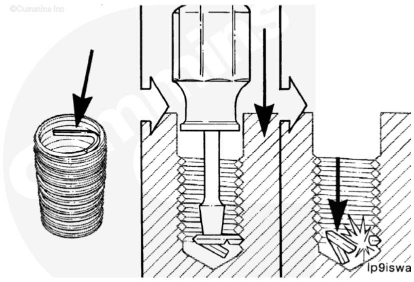
Select the type of installation tool required for your specific thread insert and space limitations.



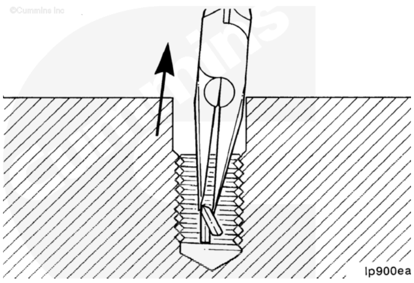
Install the repair insert into the prepared hole. The top of the insert must be 1/4 to 1/2 of a thread below the surface of the part.



Break the installation tang off of the thread repair insert. Position a punch or screwdriver near the free end of the installation tang. Strike the tool to break off the tang.



Remove the installation tang from the repaired hole.



Coiled Thread Insert Part Numbers - Metric Threads			
Thread Size	**Thread Length - mm**	**Heli-Coil™ Part Number**	**Perma Thread™ Part Number**
M6 X 1.00	9	1084-6CN-0070	206M-8C-0090
M6 X 1.00	12	1084-6CN-0120	206M-6C-0120
M6 X 1.25	12	1084-8CN-0120	206M-8C-0120
M6 X 1.25	16	1084-8CN-0160	206M-8C-0160
M10 X 1.50	15	1084-10CN-0150	206M-10C-0150
M10 X 1.50	20	1084-10CN-0200	206M-10C-0200
M12 X 1.75	18	1084-12CN-0180	206M-12C-0180
M12 X 1.75	24	1084-12CN-0240	206M-12C-0240
M14 X 2.00	21	1084-14CN-0210	206M-14C-0210
M14 X 2.00	28	1084-14CN-0280	206M-14C-0280
M16 X 2.00	24	1084-16CN-0240	206M-16C-0240
M16 X 2.00	38	1084-16CN-0320	206M-16C-0320

Drill Sizes				
Thread Size	**Suggested Drill Size - Metric (mm)**		**Suggested Drill Size - U.S. Customary [Inch]**	
	Aluminum	**Steel**	**Aluminum**	**Steel**
M6 X 1.00	6.25	6.30	D [0.2460]	1/4 [0.2500]
M8 X 1.25	8.30	8.40	21/64 [0.3281]	Q [0.3320]
M10 X 1.50	10.5	10.5	Z [0.4130]	Z [0.4130]
M12 X 1.75	12.5	12.5	31/64 [0.4844]	1/2 [0.500]
M14 X 2.00	14.5	14.5	37/64 [0.5781]	37/64 [0.5781]
M16 X 2.00	16.5	16.5	21/32 [0.6562]	21/32 [0.6562]

Tap Part Numbers				
Thread Size	**Heli-Coil™ Plug Tap Part Number**	**Heli-Coil™ Bottoming Tap Part Number**	**Perma-Thread™ Plug Tap Part Number**	**Perma-Thread™ Bottoming Tap Part Number**
M6 X 1.00	2087-6	2093-6	5043-060CP5	5043-060CB5
M8 X 1.25	2087-8	2093-8	5043-080CP5	5043-080CB5
M10 X 1.50	2087-10	2093-10	5043-100CP5	5043-100CB5
M12 X 1.75	2087-12	2093-12	5043-120CP5	5043-120CB5
M14 X 2.00	2087-14	2093-14	5043-140CP5	5043-140CB5
M16 X 2.00	2087-16	2093-16	5043-160CP5	5043-160CB5

Coiled Thread Insert Part Numbers - Unified Threads					
Thread Size and Pitch	**Thread Length-Inch**	**Cummins® Part Number**	**Heli-Coil™ Part Number**	**Perma-Thread™ Part Number**	**Coil Thread Part Number**
1/4-20	0.500	113759	1185-4CN-0500	208C4-0500	TNC-4C-0500
5/16-18	0.625	68372	1185-5CN-0625	208C5-0625	TNC-5C-0625
5/16-24	0.625	69003	1191-5CN-0625	208F5-0625	TNF-5C-0625
3/8-16	0.563	67184-A	1185-6CN-0563	208C6-0750	TNC-6C-0562
3/8-16	0.750	67184-B	1185-6CN-0750	208C6-0750	TNC-6C-0750
3/8-16	0.938	67184-C	1185-6CN-0938	208C6-0938	TNC-6C-0938
3/8-16	1.125	67184-D	1185-6CN-1125	208C6-1125	TNC-6C-1125
3/8-24	0.375	67235-A	1191-6CN-0375	208F6-0375	TNF-6C-0375
3/8-24	0.563	67235-B	1191-6CN-0563	208F6-0562	TNF-6C-0562
3/8-24	0.750	67235-C	1191-6CN-0750	208F6-0750	TNF-6C-0750
3/8-24	0.938	67235-D	1191-6CN-0938	208F6-0938	TNF-6C-0938
3/8-24	1.125	67235-E	1191-6CN-1125	208F6-1125	TNF-6C-1125
7/16-14	0.656	67234-A	1185-7CN-0656	208C7-0656	TNC-7C-0656
7/16-14	0.875	67234-B	1185-7CN-0875	208C7-0875	TNC-7C-0875
7/16-14	1.094	67234-C	1185-7CN-1094	208C7-1094	TNC-C7-1094
7/16-14	1.312	67234-D	1185-7CN-1312	208C7-1312	TNC-7C-1312
1/2-13	0.750	67183-A	1185-8CN-0750	208C8-0750	TNC-8C-0750
1/2-13	1.000	67183-B	1185-8CN-1000	208-C8-1000	TNC-8C-1000

Coiled Thread Insert Part Numbers - Unified Threads					
Thread Size and Pitch	**Thread Length-Inch**	**Cummins® Part Number**	**Heli-Coil™ Part Number**	**Perma-Thread™ Part Number**	**Coil Thread Part Number**
1/2-13	1.250	37183-C	1185-8CN-1250	208C8-1250	TNC-8C-1250
1/2-13	1.500	67183-D	1185-8CN-1500	208C8-1500	TNC-8C-1500
9/16-18	1.125	69070	1191-9CN-1125	208F9-1125	TNF-9C-1125
5/8-11	0.875	101541	1185-10CN-0875	208C10-0875	
5/8-11	1.250	101540	1185-10CN-1250	208C10-1250	TNC-10C-0938
5/8-18	0.938	163398	1191-10CN-0938	208F10-0938	TNF-10C-0938
3/4-10	1.125	101542	1185-12CN-1125	208C12-1125	TNC-12C-1125
3/4-16	1.500	132672	1191-12CN-1500	208C12-1500	TNF-12C-1500
7/8-14	0.875	100292	1191-14CN-0875	208F14-0875	TNF-14C-0875
7/8-14	1.312	100854	1191-14CN-1312	208F14-1312	TNF-14C-1312
1-8	0.680	200845	389-16CN-0680	1328-16-0680	
1-1/8-7	2.250	102588	1185-18CN-2250	208C18-2250	TNC-18C-2250

Drill Size and "Plug" Tap Part Numbers - Unified Threads					
Nominal Thread Size and Pitch	**Tap Drill Size - Inch Aluminum**	**Cast Iron**	**Heli-Coil™ Part Number**	**Coil Thread Part Number**	**Perma-Thread™ Part Number**
1/4-20	H [0.2660]	H [0.2660]	4CPA H3	CTC-4SRP-H3	1043-4CP2-H3
5/16-18	Q [0.3320]	Q [0.3320]	5CPA H4	CTC-5SRP-H4	1043-5CP2-H4
5/16-24	21/64 [0.3281]	21/64 [0.3281]	5FPA H2	CTF-5SRP-H3	1043-5FP2-H3
3/8-16	X [0.3970]	X [0.3970]	6CPA H4	CTC-6SRP-H4	1043-6CP2-H4

Drill Size and "Plug" Tap Part Numbers - Unified Threads					
Nominal Thread Size and Pitch	**Tap Drill Size - Inch Aluminum**	**Cast Iron**	**Heli-Coil™ Part Number**	**Coil Thread Part Number**	**Perma-Thread™ Part Number**
3/8-24	25/64 [0.3906]	25/64 [0.3906]	6FPA H3	CTF-6SRP-H3	1043-6FP2-H3
7/16-14	29/64 [0.4531]	29/64 [0.4531]	7CPA H4	CTC-7SRP-H4	1043-7CP2-H4
1/2-13	33/64 [0.5156]	29/64 [0.5312]	8CPA H4	CTC-8SRP-H4	1043-8CP2-H4
9/16-18	37/64 [0.5781]	37/64 [0.5781]	18193-9 H4	CTF-9SRP-H4	1043-9FP2-H4
5/8-11	21/32 [0.6562]	21/32 [0.6562]	18187-10 H4	CTC-10SRP-H4	1043-10CP2-H4
5/8-18	41/64 [0.7656]	41/64 [0.7656]	18193-10 H4	CTC-10SRP-H4	1043-10CP2-H4
3/4-10	25/32 [0.7812]	25/32 [0.7812]	18187-12 H5	CTC-12SRP-H5	1043-12CP2-H5
3/4-16	49/64 [0.7656]	49/64 [0.7656]	18193-12 h4	CTF-12SRP-H4	1043-12FP2-H4
7/8-14	57/64 [0.8906]	49/64 [0.8906]	18193-14 H4	CTF-14SRP-H4	1043-14FP2-H4
1-8	1-1/32 [1.0312]	1-1/32 [1.0312]	18187-16 H6	CTC-16SRP-H6	1043-16CP2-H6
1-1/8-7	1-11/64 [1.1719]	1-11/64 [1.1719]	18187-18 H6	CTC-18SRP-H6*	1043-18CP2-H6*
* These taps are for aluminum and magnesium metals **only**. **Note:** All taps are class 2B unless otherwise specified.					

Drill Size and Bottom Tap Part Numbers - Unified Threads					
Nominal Thread Size and Pitch	**Tap Drill Size - Inch Aluminum**	**Cast Iron**	**Heli-Coil™ Part Number**	**Coil Thread Part Number**	**Perma-Thread™ Part Number**
1/4-20	H [0.2660]	H [0.2660]	4CBA H3	CTC-4SRB-H3	1043-4CB2-H3
5/16-18	Q [0.3320]	Q [0.3320]	5CBA H4	CTC-5SRB-H4	1043-5CB2-H4
5/16-24	21/64 [0.3281]	21/64 [0.3281]	5FBA H2	CTF-5SRB-H3	1043-5FB2-H3
3/8-16	X [0.3970]	X [0.3970]	6CBA H4	CTC-6SRB-H4	1043-6CB2-H4

Drill Size and Bottom Tap Part Numbers - Unified Threads

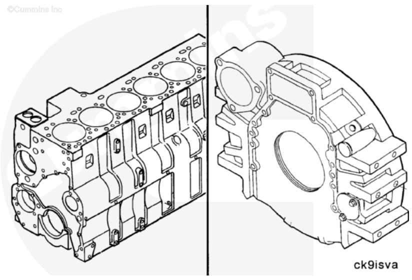
Nominal Thread Size and Pitch	Tap Drill Size - Inch Aluminum	Cast Iron	Heli-Coil™ Part Number	Coil Thread Part Number	Perma-Thread™ Part Number
3/8-24	25/64 [0.3906]	25/64 [0.3906]	6FBA H3	CTF-6SRB-H3	1043-6FB2-H3
7/16-14	29/64 [0.4531]	29/64 [0.4531]	7CBA H4	CTC-7SRB-H4	1043-7CB2-H4
1/2-13	33/64 [0.5156]	29/64 [0.5312]	8CBA H4	CTC-8SRB-H4	1043-8CB2-H4
9/16-18	37/64 [0.5781]	37/64 [0.5781]	20193-9 H4	CTF-9SRB-H4	1043-9FB2-H4
5/8-11	21/32 [0.6562]	21/32 [0.6562]	20187-10 H4	CTC-10SRB-H4	1043-10CB2-H4
5/8-18	41/64 [0.7656]	41/64 [0.7656]	20193-10 H4	CTC-10SRB-H4	1043-10CB2-H4
3/4-10	25/32 [0.7812]	25/32 [0.7812]	20187-12 H5	CTC-12SRB-H5	1043-12CB2-H5
3/4-16	49/64 [0.7656]	49/64 [0.7656]	20193-12 h4	CTF-12SRB-H4	1043-12FB2-H4
7/8-14	57/64 [0.8906]	49/64 [0.8906]	20193-14 H4	CTF-14SRB-H4	1043-14FB2-H4
1-8	1-1/32 [1.0312]	1-1/32 [1.0312]	20187-16 H6	CTC-16SRB-H6	1043-16CB2-H6
1-1/8-7	1-11/64 [1.1719]	1-11/64 [1.1719]	20187-18 H6	CTC-18SRB-H6*	1043-18CB2-H6*

* These taps are for aluminum and magnesium metals **only**.

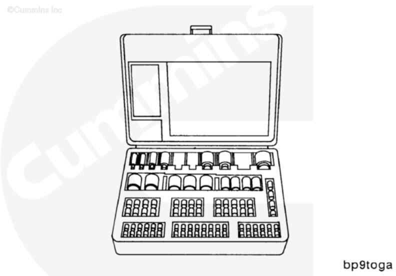
Note: All taps are class 2B unless otherwise specified.

Note : This section of the procedure provides guidelines for using keylock inserts.

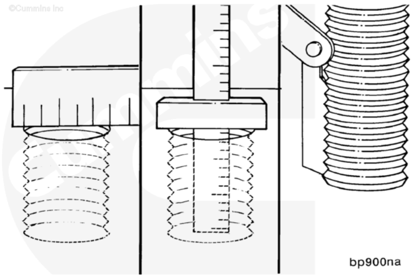
Keylock inserts can be used to repair threads in cast iron, steel, and aluminum components.



Thread repair kit, Part Number 3375021, contains keylock inserts. The inserts are internally and externally threaded metallic cylinders which have captive locking keys. The keys prevent the insert from rotating after it has been installed.

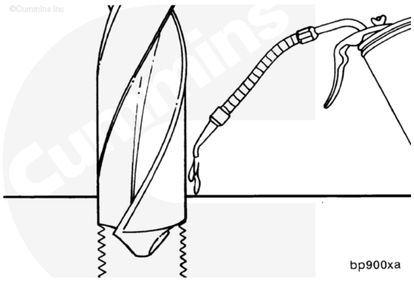


Determine the diameter, depth, and pitch of the original threaded hole.



⚠ WARNING ⚠

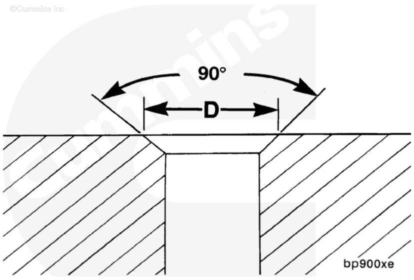
To reduce the possibility of personal injury, always wear protective goggles when drilling metal.



Use cutting oil while drilling the capscrew hole to the specified diameter. Use the charts in this procedure for drill size diameter specifications. Drill to the bottom of blind capscrew holes.

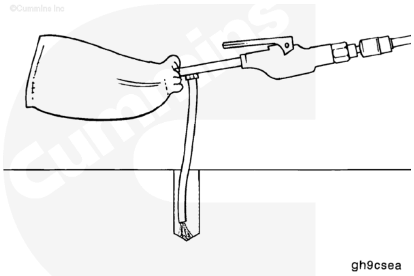
Note : The drill must be in proper alignment when drilling the capscrew hole.

Countersink the hole to the diameter specified on the chart. The correct angle is 82 to 100 degrees.



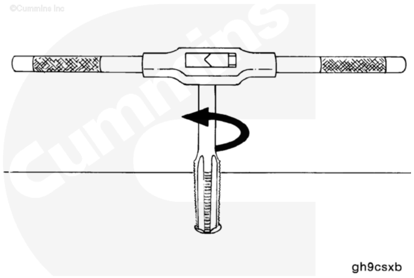
Note : Use cutting oil when countersinking the capscrew hole.

Use chip removal unit, Part Number 3164019, to remove any debris from the hole.

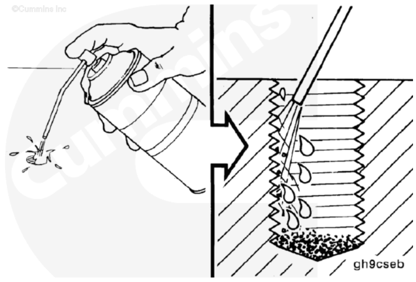


Use cutting oil to tap the hole to the correct diameter, pitch, and depth.

Tap size and depth information is included at the end of this procedure.

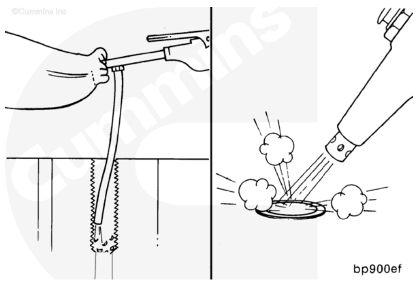


Use a spray cleaner to clean the oil from the capscrew hole and flush the newly cut threads.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

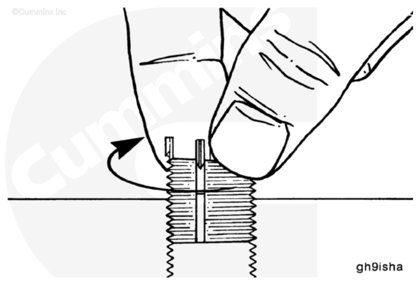


Use chip removal unit, Part Number 3164019, to remove any debris from the hole.

Dry the threads with compressed air.

Install the insert into the prepared hole.

Turn the insert **clockwise** until the insert stops at the specified depth.

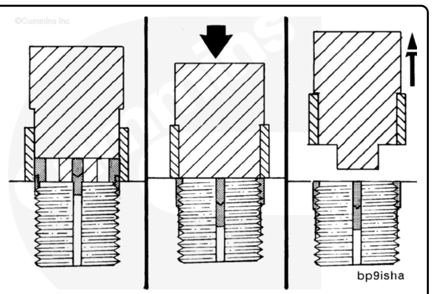


Put the staking tool over the insert. Slide the sleeve of the tool over the keys.

Use a hammer to strike the installation tool until the keys are driven down to their full depth.

Remove the tool.

The repair is now complete.



Keylock Thread Inserts - Unified Threads							
Cummins® Part Number	Internal Thread Size Class 2B	Tap Drill Size - Inch	Tap Size - Inch Class 2B	Installation Tool Part Number	Minimum Tap Depth - Inch	Countersink Diameter - Inch	Overall Length - Inch
3375141	1/4-20	0.332	3/8-16	3375142	0.43	0.385	0.37
3375022	5/16-18	0.397	7/16-14	3375023	0.50	0.447	0.43
3375430	5/16-18 (Heavy Duty)	0.453	1/2-13	3375026	0.56	0.510	0.43
3375024	3/8-16	0.453	1/2-13	3375026	0.56	0.510	0.50
3375025	3/8-24	0.453	1/2-13	3375026	0.56	0.510	0.50
3375027	7/16-14	0.516	9/16-12	3375029	0.62	0.572	0.56
3375028	7/16-20	0.516	9/16-12	3375029	0.62	0.572	0.56
3375030	1/2-13	0.578	5/8-11	3375032	0.68	0.635	0.62
3375031	1/2-20	0.578	5/8-11	3375032	0.68	0.635	0.62
3375143	5/8-11	0.766	13/16-16	3375034	0.87	0.822	0.81
3375033	5/8-18	0.766	13/16-16	3375034	0.87	0.822	0.81
3375035	3/4-16	1.062	1-1/8-12	3375036	1.18	1.145	1.12
3375037	1-18	1.312	1-3/8-12	3375038	1.56	1.395	1.37

Keylock Thread Inserts - Tridair Keenserts™ - Unified Threads							
Part Number	Internal Thread Size Class 2B	External Thread Size Class 2A	Length Inch	Tap Drill Diameter	Countersink Diameter	Tap Size Class 2 B	Insert Tool Part Number
RKKA 1/4-20	1/4-20	3/8-16	0.37	"Q"	0.38	3/8-16	TRKA 1/4
RKKA 5/16-18	5/16-18	7/16-14	0.43	"X"	0.44	7/16-14	TRKA 5/16
RKK 5/16-18 (Heavy Duty)	5/16-18	1/2-13	0.43	29/64	0.51	1/2-13	TRK 5/16
RKKA 3/8-16	3/8-16	1/2-13	0.50	29/64	0.51	1/2-13	TRKA 3/8
RKKA 3/8-24	3/8-24	1/2-13	0.50	29/64	0.51	1/2-13	TRKA 3/8
RKKA 7/16-14	7/16-14	9/16-12	0.56	33/64	0.57	9/16-12	TRKA 7/16

Keylock Thread Inserts - Tridair Keenserts™ - Unified Threads							
Part Number	**Internal Thread Size Class 2B**	**External Thread Size Class 2A**	**Length Inch**	**Tap Drill Diameter**	**Countersink Diameter**	**Tap Size Class 2 B**	**Insert Tool Part Number**
RKKA 7/16-20	7/16-20	9/16-12	0.56	33/64	0.57	9/16-12	TRKA 7/16
RKKA 1/2-13	1/2-13	5/8-11	0.62	37/64	0.63	5/8-11	TRKA 1/2
RKKA 1/2-20	1/2-20	5/8-11	0.62	37/64	0.63	5/8-11	TRKA 1/2
RKK 5/8-11	5/8-11	7/8-14	0.87	53/64	0.88	7/8-14	TRK 5/8
RKK 5/8-18	5/8-18	7/8-14	0.87	53/64	0.88	7/8-14	TRK 5/8
RKK 3/4-10	3/4-10	1-1/8-12	1.12	1-1/16	1.14	1-1/8-12	TRK 3/4
RKK 3/4-16	3/4-16	1-1/8-12	1.12	1-1/16	1.14	1-1/8-12	TRK 3/4

Keylock Thread Inserts - Tridair Keenserts™ - Metric Threads							
Part Number	**Internal Thread Size**	**External Thread Size**	**Length - mm**	**Tap Drill Diameter**	**Countersink Diameter**	**Tap Size**	**Insert Tool Part Number**
KNM 6x1F	M6 x 1.0	M10 x 1.25	10	8.8	10.25	M10 x 1.25	TRKM 6
KNM 8x1.25F	M8 x 1.25	M12 x 1.25	12	10.8	12.25	M12 x 1.25	TRKM 8
KNM 10x1.5F	M10 x 1.50	M14 x 1.5	14	12.8	14.25	M14 x 1.5	TRKM 10
KNM 12x1.75 F	M12 x 1.75	M16 x 1.5	16	14.75	16.25	M16 x 1.5	TRKM 12
KNHM 14x2F	M14 x 2.0	M20 x 1.5	20	18.75	20.25	M20 x 1.5	TRKHM 14

Keylock Thread Inserts - Microdot K-Serts - Unified Threads							
Part Number	**Internal Thread Size Class 2B**	**External Thread Size Class 2A**	**Length - inch**	**Tap Crill Diameter**	**Countersink Diameter**	**Tap Size Class 2B**	**Insert Tool Part Number**
K51240-420	1/4-20	3/8-16	0.37	0.331-0.336	0.39	3/8-16	KT151240-4

Keylock Thread Inserts - Microdot K-Serts - Unified Threads							
Part Number	**Internal Thread Size Class 2B**	**External Thread Size Class 2A**	**Length - inch**	**Tap Crill Diameter**	**Countersink Diameter**	**Tap Size Class 2B**	**Insert Tool Part Number**
K51240-518	5/16-18	7/16-14	0.43	0.376-0.401	0.45	7/16-14	KT151240-5
K51241-518 (Heavy Duty)	5/16-18	1/2-13	0.43	29/64	0.51	1/2-13	KT151241-5
K51240-616	3/8-16	1/2-13	0.50	29/64	0.51	1/2-13	KT151240-6
K51240-6	3/8-24	1/2-13	0.50	29/64	0.51	1/2-13	KT151240-6
K51240-714	7/16-14	9/16-12	0.56	33/64	0.57	9/16-12	KT51240-7
K51240-7	7/16-20	9/16-12	0.56	33/64	0.57	9/16-12	KT51240-7
K51240-813	1/2-13	5/8-11	0.62	37/64	0.64	5/8-11	KT51240-8
K51240-8	1/2-20	5/8-11	0.62	37/64	0.64	5/8-11	KT51240-8
K51241-1011	5/8-11	7/8-14	0.87	53/64	0.89	7/8-14	KT51241-10
K51241-10	5/8-18	7/8-14	0.87	53/64	0.89	7/8-14	KT51241-10
K51241-1210	3/4-10	1-1/8-12	1.12	1-1/16	1.15	1-18-12	KT51241-12
K51241-12	3/4-16	1-1/8-12	1.12	1-1/16	1.15	1-1/8-12	KT51241-12

Keylock Thread Inserts - Microdot K-serts - Metric Threads							
Part Number	**Internal Thread Size**	**External Thread Size**	**Length - mm**	**Tap Drill Diameter**	**Countersink Diameter**	**Tap Size**	**Insert Tool Part Number**
KM9900-060	M6 x 1.00	M10 x 1.25	10.0	8.8	10.25	M10 x 1.25	KTM9000-060
KM9900-080	M8 x 1.25	M12 x 1.25	12.0	10.8	12.25	M12 x 1.25	KTM9000-080
KM9900-100	M10 x 1.50	M14 x 1.50	14.0	12.8	14.25	M14 x 1.50	KTM9000-100

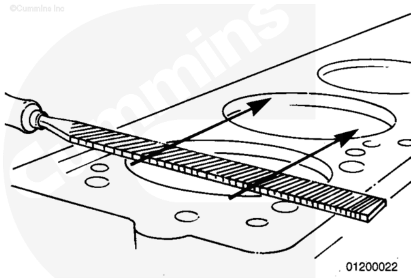
Keylock Thread Inserts - Microdot K-serts - Metric Threads							
Part Number	**Internal Thread Size**	**External Thread Size**	**Length - mm**	**Tap Drill Diameter**	**Countersink Diameter**	**Tap Size**	**Insert Tool Part Number**
KM9900-120	M12 x 1.75	M16 x 1.50	16.0	14.75	16.25	M16 x 1.50	KTM9000-120
KHM9900-140	M14 x 2.00	M20 x 1.50	20.0	18.75	20.25	M20 x 1.50	KHTM9000-140
KHM9900-160	M16 x 2.00	M22 x 1.50	22.0	20.50	22.25	M22 x 1.50	KHTM9000-160

Note : This section of the procedure is for Open End Thread Repair Inserts.

This procedure describes the method recommended for repairing damaged metric threads. Installing metric open end thread repair inserts is an approved method for warranty repairs and can be performed without removing the engine from the chassis. Use care to prevent contaminants from entering the engine during this process.

Inspect the area around the threaded hole that is to be repaired.

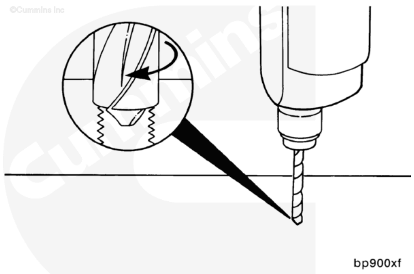
Remove any burrs from the surface. A flat mill file is very effective for this process. Burr removal is necessary so an accurate fixture location can be obtained.



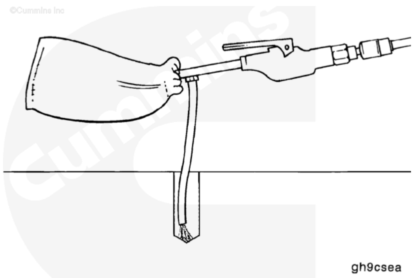
⚠ WARNING ⚠

To reduce the possibility of personal injury, always wear protective goggles when drilling metal.

Use a cutting oil when drilling the damaged threads to the specified diameter and depth dimensions for the specified repair insert. A listing of installation requirements is included in this procedure.

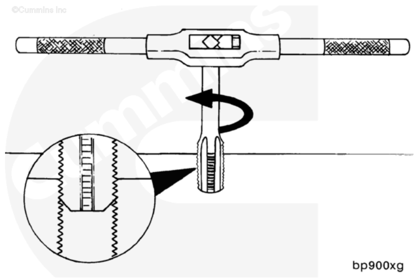


Use chip removal unit, Part Number 3164019, to remove any debris from the hole.



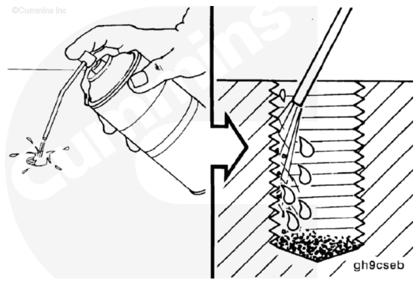
Use cutting oil. Tap the hole for the repair insert. The correct thread depth will allow the insert to be installed 0.0 to 0.25 mm [0.00 to 0.010 inch] below the surface of the component.

Measure the insert and compare the measurement to the data sheet contained in this procedure for insert length specifications.



Use a spray cleaner to clean the oil from the capscrew hole.

Flush the newly cut threads.



⚠ WARNING ⚠

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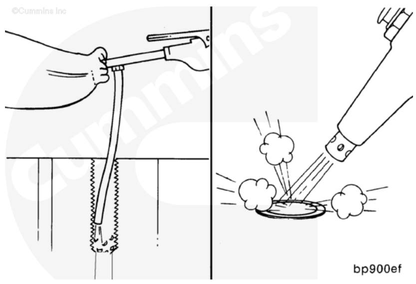
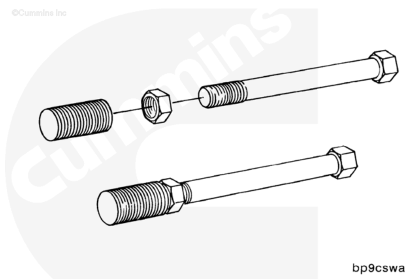


Use chip removal unit, Part Number 3164019, to remove any debris from the hole.

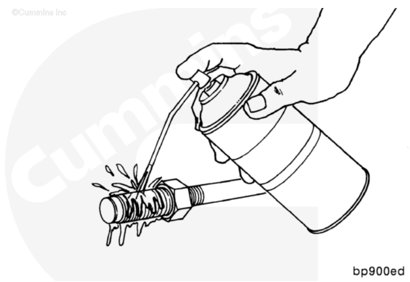
Dry the threads with compressed air.

Obtain a capscrew and nut with the same size threads as the insert inside diameter.

Install the nut and insert on the capscrew.

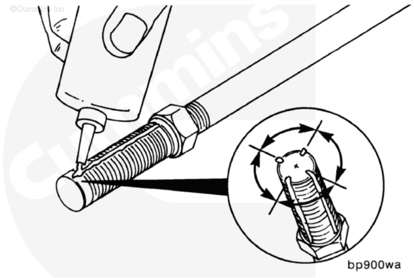


Use a spray cleaner to remove the preservative coating from the outside diameter of the insert.

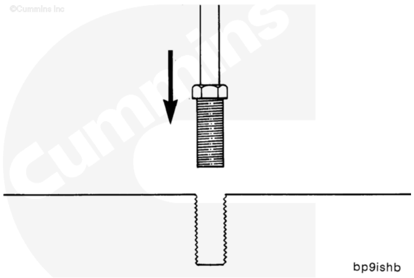


Apply four beads of sealant, Part Number 3375068, to the outside diameter of the insert.

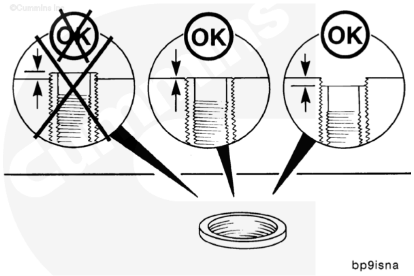
Note : The beads of sealant must be about 0.8 mm [1/32 inch] wide and spaced 90 degrees from each other.



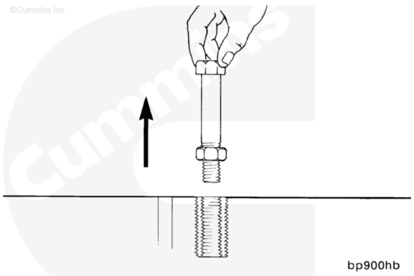
Install the repair insert.



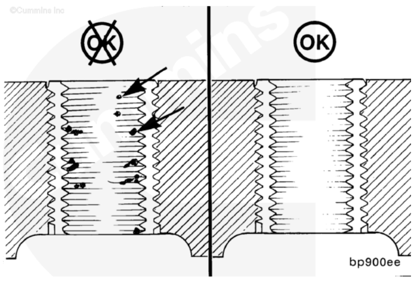
The insert **must** be 0.0 to 0.25 mm [0.00 to 0.010 inch] below the surface.



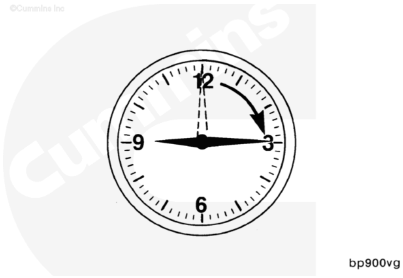
Loosen the nut and remove the capscrew.



Make sure the inside portion of the repair insert is clean. Remove any file shavings or debris.

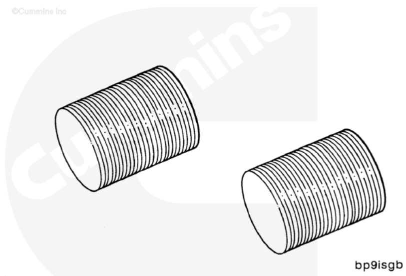


Allow the sealant to cure for 15 minutes before use.



Reference Data and Functional Checklist				
Cummins® Part Number	**Internal Thread (Size it repairs)**	**Overall Length - mm**	**Tap Drill Diameter - mm**	**External Thread (Tap Size)**
3376701	M6 x 1.00-6H	12.0	8.8	M10 x 1.25-6g
3376702	M8 x 1.25-6H	12.0	10.8	M12 x 1.25-6g
3376703	M8 x 1.25-6H	17.0	10.8	M12 x 1.25-6g
3376704	M8 x 1.25-6H	20.0	10.8	M12 x 1.25-6g
3376705	M10 x 1.50-6H	08.0	12.8	M14 x 1.50-6g
3376706	M10 x 1.50-6H	12.5	12.8	M14 x 1.50-6g
3376707	M10 x 1.50-6H	14.0	12.8	M14 x 1.50-6g
3376709	M10 x 1.50-6H	18.0	12.8	M14 x 1.50-6g
3376710	M10 x 1.50-6H	20.0	12.8	M14 x 1.50-6g
3376711	M10 x 1.50-6H	24.0	12.8	M14 x 1.50-6g
3376712	M12 x 1.25-6H	08.0	14.75	M16 x 1.50-6g
3376713	M12 x 1.75-6H	08.0	14.75	M16 x 1.50-6g
3376714	M12 x 1.75-6H	21.0	14.75	M16 x 1.50-6g
3376715	M14 x 2.00-6H	30.0	16.5	M18 x 1.50-6g
3376716	M16 x 2.00-6H	30.0	18.0	M20 x 2.00-06g

Note : This part of this procedure applies to Closed End "Blind" Inserts.



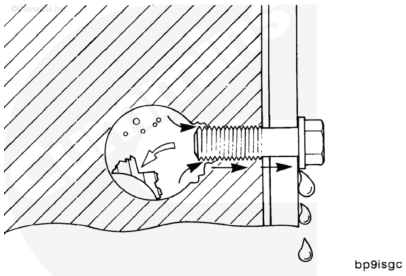
This part of the procedure provides the instructions for installing closed end "blind" inserts.

The charts contained in this procedure lists the two types of blind inserts.

- SAE internal and external threads
- Metric internal with SAE external threads.

Closed end inserts are generally used in cases where the bottom or side of a hole has been pierced. They can be used to repair any kind of hole or thread, if desired.

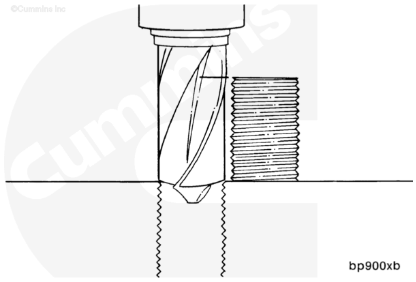
Note : Closed end inserts can not be used to repair top deck cracks that run into cylinder head capscrew holes.



Obtain the required reamer or drill size for the hole to be repaired. Use the charts contained in this procedure for size specifications. Install it in the drill.

Set the drill depth by placing the insert to be installed between the top of the hole and the drill chuck.

Note : If the drill is too long, make a mark on the drill bit to indicate the correct depth.

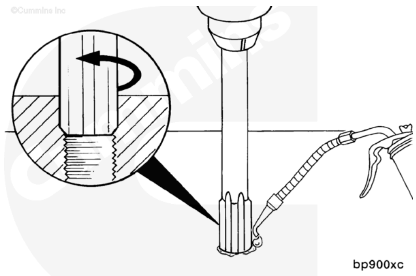


⚠ WARNING ⚠

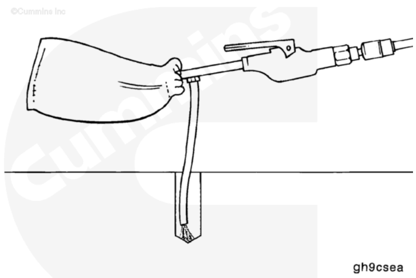
To reduce the possibility of personal injury, always wear protective goggles when drilling metal.

Use cutting oil. Drill or ream the hole to the proper depth.

Note : In some cases when using a drill, the proper depth will pierce the bottom of the hole.

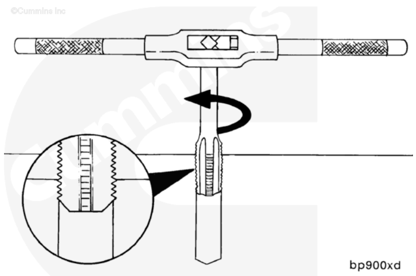


Use chip removing unit, Part Number 3164019, to remove machining chips from the capscrew hole.

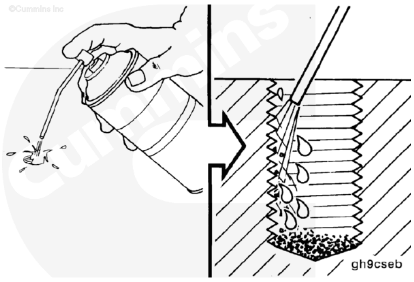


Use cutting oil. Tap the hole to the required depth. Use the charts contained in this procedure for tap sizes.

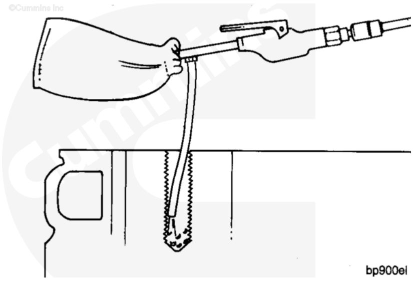
Note : The correct depth will allow the insert to be installed flush, to slightly below the surface.



Use a spray cleaner to remove the oil film and clean the newly cut threads.



Use chip removal unit, Part Number 3164019, to remove any debris from the hole.



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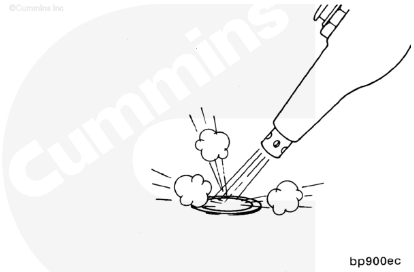
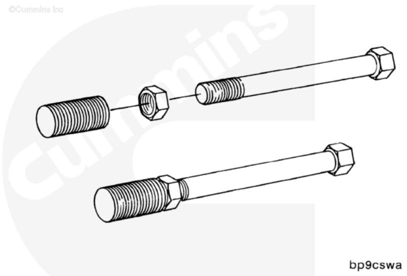


Use compressed air to dry the threads.

Obtain a capscrew and nut with the same size threads as the insert inside diameter.

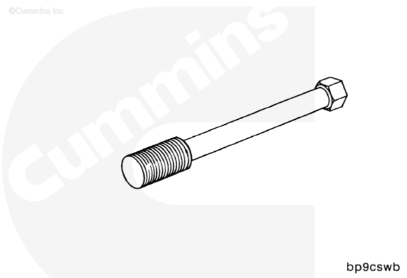
Install the nut first, then the insert on the capscrew.

Note : The nut should have a straight shoulder which, when installed against the insert, will allow the insert to be installed slightly below the surface.

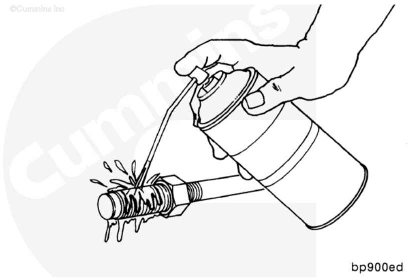


If a nut is **not** available, it is acceptable to bottom the capscrew out in the insert for installation.

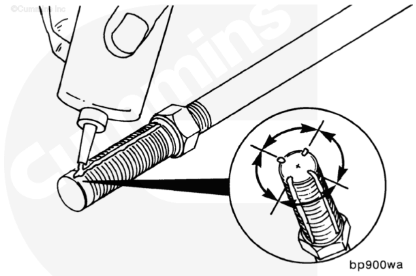
Note : This type installation will require time for the sealant to set up before removing the capscrew from the thread insert.



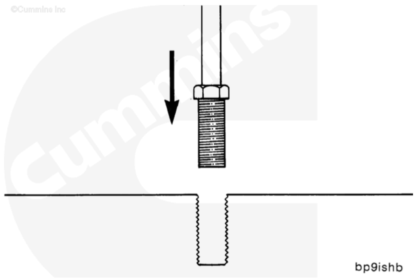
Use a spray cleaner to remove the preservative coating from the outside diameter of the insert.



Apply four beads of sealant, Part Number 3375068, to the outside diameter of the insert. The beads of sealant **must** be about 0.8 mm [1/32 inch] wide and spaced 90 degrees from each other.

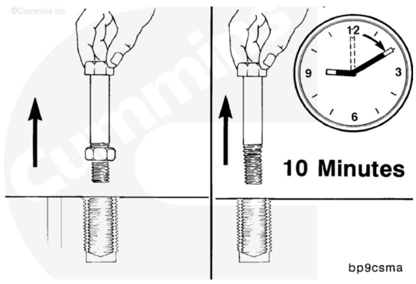


Install the repair insert.

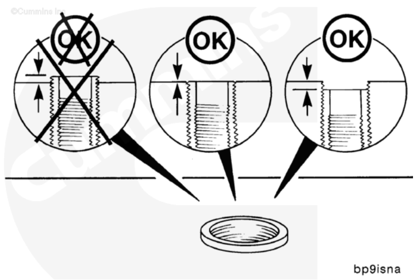


Loosen the nut and remove the capscrew.

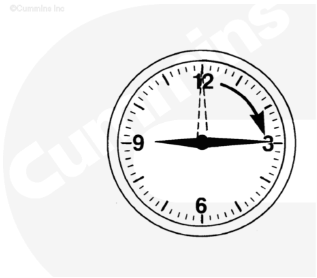
Note : If a capscrew without a nut was used, allow 10 minutes for the sealant to cure before removing the capscrew.



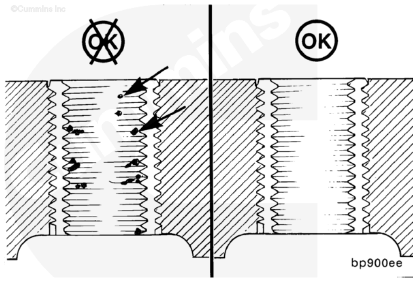
The insert should be flush to slightly below the surface.



Allow the sealant to cure for 15 minutes before use.



Clean the insert threads to remove any debris.



Closed End Inserts - SAE Internal and External Threads				
Cummins® Part Number	**Internal Thread (Size it repairs) Class 2B**	**External Threads (Tap Size) Class 2B**	**Tap Drill (Reamer) Diameter - Inch**	**Nominal Overall Length - Inch**
3376437	1/4-20 UNC	3/8-16 UNC	0.332	0.750
3376438	5/16-18 UNC	7/16-14 UNC	0.397	0.930
3376439	3/8-16 UNC	1/2-13 UNC	0.453	1.130
3376440	3/8-24 UNF	1/2-13 UNC	0.453	1.070
3376441	7/16-14 UNC	9/16-12 UNC	0.516	1.150
3376442	7/16-20 UNF	9/16-12 UNC	0.516	1.210
3376443	1/2-13 UNC	5/8-11 UNC	0.578	1.510
3376446	1/2-20 UNF	5/8-11 UNC	0.578	1.190
3376444	5/8-11 UNC	13/16-16 UNC	0.766	1.640
3376445	5/8-18 UNF	13/16-16 UNC	0.766	1.360
3376447	9/16-18 UNF	11/16-11 UNC	0.641	1.180
Closed End Inserts - Metric Internal Threads, SAE External Threads (3822709 KIT)				
Cummins® Part Number	**Internal Threads (Size it repairs)**	**External Threads (Tap Size) Class 2B**	**Tap Drill (Reamer) Daimeter - Inch**	**Nominal Overall Length - Inch**
3822698	M 6X 1.0-6H	3/8-16 UNC	0.3125	0.75
3822699	M 8X 1.25-6H	1/2-13 UNC	0.421	0.94
3822700	M 10X 1.50-6H	9/16-12 UNC	0.484	1.12
3822701	M 12X 1.25-6H	5/8-18 UNF	0.578	1.37
3822702	M 12X 1.50-6H	5/8-18 UNF	0.578	1.37

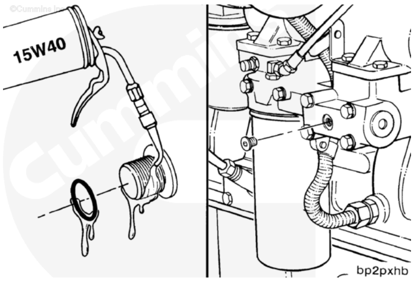
Closed End Inserts - Metric Internal Threads, SAE External Threads (3822709 KIT)				
Cummins® Part Number	**Internal Threads (Size it repairs)**	**External Threads (Tap Size) Class 2B**	**Tap Drill (Reamer) Daimeter - Inch**	**Nominal Overall Length - Inch**
3822703	M 12X 1.75-6H	5/8-18 UNF	0.578	1.37
3822704	M 14X 2.00-6H	3/4-16 UNF	0.687	1.50
3822705	M 16X 1.50-6H	7/8-14 UNF	0.812	1.75
3822706	M 16X 2.00-6H	7/8-14 UNF	0.812	1.75

Install

Install a new o-ring on the plug. Lubricate with clean 15W-40 oil.

Install and tighten the plugs.

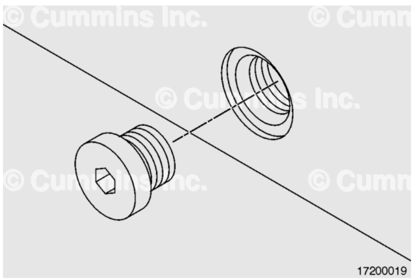
Use the following chart for torque values.



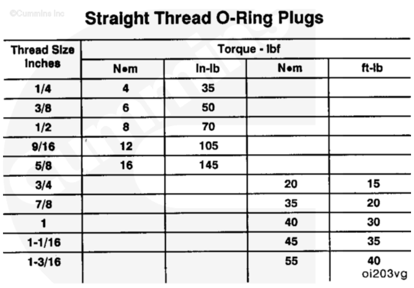
Install a new o-ring on the plug. Lubricate the o-ring with clean 15W-40 oil.

Install and tighten the plugs.

Use to the following chart for torque values.



Tighten straight thread plugs to the appropriate torque value.



Straight Thread O-Ring Plugs

Thread Size Inches	Torque - lbf			
	N•m	In-lb	N•m	ft-lb
1/4	4	35		
3/8	6	50		
1/2	8	70		
9/16	12	105		
5/8	16	145		
3/4			20	15
7/8			35	20
1			40	30
1-1/16			45	35
1-3/16			55	40

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